



School of Computer Science and Engineering
Winter Semester 2023-2024

Continuous Assessment Test – II
SLOT: A2+TA2

Programme Name & Branch : B.Tech (BCB/BCE/BCI/BCT/BDS/BKT)

Course Name & code: Design and Analysis of Algorithms, BCSE204L

Class Number (s): ALL

Faculty Name (s): ALL

Exam Duration: 90 Min.

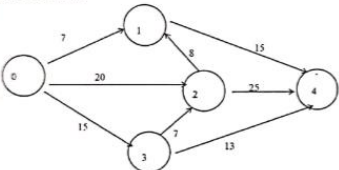
Maximum Marks: 50

Answer All

Q.No.	Question	Max Marks															
1.	A bank has 14 million dollars, which can be invested into stocks of four companies (1, 2, 3, and 4). The following table shows the net revenue of each company and the amount that must be invested into each company. <table><tr><th>Company</th><th>1</th><th>2</th><th>3</th><th>4</th></tr><tr><td>Revenue (Million \$)</td><td>16</td><td>22</td><td>12</td><td>8</td></tr><tr><td>Investment Amount (Million \$)</td><td>5</td><td>7</td><td>4</td><td>3</td></tr></table> The objective for the bank is to select a set of companies for investment, so as to maximize the total revenue with the condition that no partial investment can be done i.e., for each company we can either invest into it or not. Solve the problem with the algorithm whose exponential worst-case time complexity can be improved by employing better techniques for efficient pruning.	Company	1	2	3	4	Revenue (Million \$)	16	22	12	8	Investment Amount (Million \$)	5	7	4	3	10
Company	1	2	3	4													
Revenue (Million \$)	16	22	12	8													
Investment Amount (Million \$)	5	7	4	3													

2.	i) Find out the longest proper prefix that is also the proper suffix for the pattern "ababa". ii) Calculate the LPS table of the pattern "ababaa". Using the table constructed in (ii), answer the following. iii) What is the LPS value for the 4th character 'b'? What does that value infer? iv) Let 'i' and 'j' be used to denote the index of the characters to be compared in text and pattern respectively. For the text "abababababab", if the comparison starts from the beginning of the text and if all the characters of the pattern matched with those of the text, what will be the new values of i and j, to proceed with the next comparison? v) For the text "abababababab", if all the characters of the pattern except the last, matched with those of the text, what will be the new values of i and j, to proceed with the next comparison?	10
3.	A) <pre>RELAX(u, v, w) 1 if u.d > u.d + w(u, v) 2 u.d = u.d + w(u, v) BELLMAN-FORD(G, w, s) 1 INITIALIZE-SINGLE-SOURCE(G, s) 2 for i = 1 to G.V - 1 3 for each edge (u, v) ∈ G.E 4 RELAX(u, v, w) 5 for each edge (u, v) ∈ G.E 6 if u.d > u.d + w(u, v) 7 return FALSE 8 return TRUE</pre> In the above Bellman-Ford algorithm, make the required modifications to merge the two loops in line numbers 2 and 5 into a single loop and to implement early termination of the loop, if no edges are relaxed in a particular iteration. B) The capacity matrix for a flow network is given below. The first augmenting path found using Edmonds-Karp algorithm for the graph is 0-1-2-4-5. Is this correct or not? Justify your answer.	5

		0	1	2	3	4	5	
	0	0	16	13	0	0	0	
	1	0	0	0	12	0	0	
	2	0	0	0	0	14	0	
	3	0	0	9	0	0	20	
	4	0	0	0	7	0	4	
	5	0	0	0	0	0	0	

4.	Apply Push-Relabel algorithm for the flow network given below to find the maximum flow that can pass from the source to sink. 	10
5.	Two line-segments intersect if and only if either (or both) of the following conditions holds: - Each segment straddles the line containing the other. - An endpoint of one segment lies on the other segment. (This condition comes from the boundary case.) <pre> SEGMENTS-INTERSECT(p_1, p_2, p_3, p_4) 1 $d_1 = \text{DIRECTION}(p_1, p_2, p_3)$ 2 $d_2 = \text{DIRECTION}(p_2, p_3, p_1)$ 3 $d_3 = \text{DIRECTION}(p_1, p_2, p_4)$ 4 $d_4 = \text{DIRECTION}(p_3, p_4, p_1)$ 5 if ($d_1 > 0$ and $d_2 < 0$) or ($d_1 < 0$ and $d_2 > 0$) and ($d_3 > 0$ and $d_4 < 0$) or ($d_3 < 0$ and $d_4 > 0$) 6 return TRUE 7 if $d_1 \neq 0$ and $\text{ON-SEGMENT}(p_3, p_1, p_2)$ 8 return TRUE 9 if $d_2 \neq 0$ and $\text{ON-SEGMENT}(p_1, p_2, p_3)$ 10 return TRUE 11 if $d_3 \neq 0$ and $\text{ON-SEGMENT}(p_1, p_3, p_2)$ 12 return TRUE 13 if $d_4 \neq 0$ and $\text{ON-SEGMENT}(p_3, p_4, p_1)$ 14 return TRUE 15 else return FALSE DIRECTION(p_i, p_j, p_k) 1 return $(p_k - p_i) \times (p_j - p_i)$ ON-SEGMENT(p_i, p_j, p_k) 1 if $\min(x_i, x_j) \leq x_k \leq \max(x_i, x_j)$ and $\min(y_i, y_j) \leq y_k \leq \max(y_i, y_j)$ 2 return TRUE 3 else return FALSE </pre>	

	Given a line-segment L1 with two end-points (2,1) and (6,5), select the other appropriate line-segment among the following to justify your answer for the statements (A), (B) and (C) given below. L2 with two end-points (3,5) and (5,2) L3 with two end-points (3,5) and (4,4) L4 with two end-points (7,6) and (8,7) L5 with two end-points (7,6) and (5,8) L6 with two end-points (5,4) and (7,3)	
	Statement A: The condition in line number 5 of the algorithm is "if ($d_1 > 0$ and $d_2 < 0$) or ($d_1 < 0$ and $d_2 > 0$) <u>and</u> ($d_3 > 0$ and $d_4 < 0$) or ($d_3 < 0$ and $d_4 > 0$)" return TRUE" Can the "and" operator which is underlined and highlighted in bold, be replaced with "or" operator to prove that one segment straddles the line containing the other?	3
	Statement B: To prove intersection, is it a must to check the condition "ON_SEGMENT(P_i, P_j, P_k)" in line numbers 7 to 14 of the algorithm?	3
	Statement C: The else-if conditions in line numbers 7 to 14 of the algorithm show that any one of the directions d_1, d_2, d_3, d_4 can be zero to prove intersection. Is there any line-segment in the list for which only one direction is zero and does not intersect with L1? Is there any line-segment in the list for which more than one direction is zero and still does not intersect with L1?	4

1.	For the following job assignment problem, apply FIFO Branch and Bound (FIFOBB) and LIFO Branch and Bound (LIFOBB) separately and deduce the corresponding optimal assignments. The objective is to assign a unique pairing between a person and a job, such that the total time taken by all the persons to complete their assigned jobs is minimum. The rows correspond to the time taken by a person for each of the available jobs. Generate separate state space trees for FIFOBB and LIFOBB, devise an appropriate lower bound function and apply it for each of the states in the trees and finally deduce the optimal assignments. <table border="1"> <thead> <tr> <th></th><th>Job 1</th><th>Job 2</th><th>Job 3</th><th>Job 4</th></tr> </thead> <tbody> <tr> <td>Person A</td><td>11</td><td>9</td><td>4</td><td>10</td></tr> <tr> <td>Person B</td><td>9</td><td>6</td><td>5</td><td>8</td></tr> <tr> <td>Person C</td><td>10</td><td>7</td><td>3</td><td>10</td></tr> <tr> <td>Person D</td><td>8</td><td>6</td><td>11</td><td>9</td></tr> </tbody> </table> Ans. Branch and Bound Lower bound (LB): take the minimum from each row. By FIFO method : Level 0 Root node (1) Start, $LB=4+5+3+6=18$ Level 1 Allocation for Person A (4 nodes) (2) A=J1 $\rightarrow LB=11+5+3+6=25$ (3) A=J2 $\rightarrow LB=9+5+3+8=25$ (4) A=J3 $\rightarrow LB=4+6+7+6=23$ (5) A=J4 $\rightarrow LB=10+5+3+6=24$ Level 2 Allocation for Person B (4 * 3 =) 12 nodes Under (2) $\rightarrow$ (6) B=J2 $\rightarrow LB=11+6+3+9=29$ Under (2) $\rightarrow$ (7) B=J3 $\rightarrow LB=11+5+7+6=29$		Job 1	Job 2	Job 3	Job 4	Person A	11	9	4	10	Person B	9	6	5	8	Person C	10	7	3	10	Person D	8	6	11	9	10	CO2	BL3
	Job 1	Job 2	Job 3	Job 4																									
Person A	11	9	4	10																									
Person B	9	6	5	8																									
Person C	10	7	3	10																									
Person D	8	6	11	9																									

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Under (2) $\rightarrow$ (8) B=J4 $\rightarrow LB=11+8+3+6=28$ Under (3) $\rightarrow$ (9) B=J1 $\rightarrow LB=9+9+3+9=30$ Under (3) $\rightarrow$ (10) B=J3 $\rightarrow LB=9+5+10+8=32$ Under (3) $\rightarrow$ (11) B=J4 $\rightarrow LB=9+8+3+8=28$ Under (4) $\rightarrow$ (12) B=J1 $\rightarrow LB=4+9+7+6=26$ Under (4) $\rightarrow$ (13) B=J2 $\rightarrow LB=4+6+10+8=28$ Under (4) $\rightarrow$ (14) B=J4 $\rightarrow LB=4+8+7+6=25$ Under (5) $\rightarrow$ (15) B=J1 $\rightarrow LB=10+9+3+6=28$ Under (5) $\rightarrow$ (16) B=J2 $\rightarrow LB=10+6+3+8=29$ Under (5) $\rightarrow$ (17) B=J3 $\rightarrow LB=10+5+7+6=28$ Similarly, Level 3 allocation for Person C (12*2=) 24 nodes And, Level 4 allocation for Person D (24*1 =) 24 nodes Pruning/killing of nodes start here, and the pruning happens at both Level 3 and Level 4 Answer : A=J3, B=J4, C=J2, D=J1, Optimal cost = 27 For LIFO, the same nodes are created and the same optimal answer is generated, but the order of creation changes, and more pruning occurs.			
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2.	Apply a string matching algorithm (which applies hashing to reduce the time complexity of operations) to find out the shift locations where the pattern (P) matches with the text (T). $T = \text{"ABDEFACEDBDE"}$ $P = \text{"ACE"}$ Demonstrate the preprocessing operations as well as the actual shifting/matching operations. Deduce the locations of spurious hits (if any) as well as the actual matches. Use $q=11$ for the modulo operation, and an octal number system to represent the alphabet in both T and P. Ans. Rabin Karp algorithm $T = \text{"ABDEFACEDBDE"}$, $P = \text{"ACE"}$ Mapping- A=0, B=1, ..., E=4, F=5 $T = \text{"01345024134"}$, $P = \text{"024"}$, $n=11$, $m=3$, $d=8$, $q=11$, $h=8^2 \bmod 11=9$ Preprocessing $i=1$ to 3, $p=0$, $t_0=0$ $p = (d \cdot p + P[i]) \bmod 11$, $t_0 = (d \cdot t_0 + T[i]) \bmod 11$ $p = (8 \cdot 0 + 0) \bmod 11 = 0 \rightarrow (8 \cdot 0 + 2) \bmod 11 = 2 \rightarrow (8 \cdot 2 + 4) \bmod 11 = 9$ $t_0 = (8 \cdot 0 + 0) \bmod 11 = 0 \rightarrow (8 \cdot 0 + 1) \bmod 11 = 1 \rightarrow (8 \cdot 1 + 3) \bmod 11 = 0$ Processing part $t(s+1) = (d(t(s) - T[s+1] \cdot h) + T[s+m+1]) \bmod 11$ $t1 = (8(0 - 0 \cdot 9) + 4) \bmod 11 = 4$	10	CO3	BL3
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$t_2 = (8(4-1*9)+5) \bmod 11 = 9$, $p=9$ but spurious hit
 $t_3 = (8(9-3*9)+0) \bmod 11 = 10$. Similarly, $t_4=3$, $t_5=9$ (actual match), $t_6=7$, $t_7=3$, $t_8=4$

3.	A network having six nodes (SABCADE), with its set of weighted edges will be provided. The objective is to find the shortest path from S to reach all the other nodes A, B, C, D and E, using an algorithm which can work with negative weighted edges. The solution should disclose the shortest distances from S to all other nodes AND the corresponding paths through which this is possible. Generate the solution by showing all the necessary intermediate steps. The order/sequence of edges and their corresponding weights are : $A \rightarrow B$ (10), $B \rightarrow C$ (30), $A \rightarrow C$ (50), $A \rightarrow D$ (30), $C \rightarrow D$ (-100), $D \rightarrow E$ (20), $S \rightarrow A$ (5), $S \rightarrow D$ (30), $E \rightarrow S$ (30) Ans. Bellman-Ford algorithm for single source shortest path Initialization <table> <tr> <th></th> <th>S</th> <th>A</th> <th>B</th> <th>C</th> <th>D</th> <th>E</th> </tr> <tr> <td>d/cost</td> <td>0</td> <td>Inf</td> <td>Inf</td> <td>Inf</td> <td>Inf</td> <td>Inf</td> </tr> <tr> <td>P</td> <td>Nil</td> <td>Nil</td> <td>Nil</td> <td>Nil</td> <td>Nil</td> <td>Nil</td> </tr> </table> Relaxing the edges given in the same order as given in the question. After round 1 <table> <tr> <th></th> <th>S</th> <th>A</th> <th>B</th> <th>C</th> <th>D</th> <th>E</th> </tr> <tr> <td>d/cost</td> <td>0</td> <td>5</td> <td>Inf</td> <td>Inf</td> <td>30</td> <td>Inf</td> </tr> <tr> <td>P</td> <td>Nil</td> <td>S</td> <td>Nil</td> <td>Nil</td> <td>S</td> <td>Nil</td> </tr> </table> After Round 2 <table> <tr> <th></th> <th>S</th> <th>A</th> <th>B</th> <th>C</th> <th>D</th> <th>E</th> </tr> <tr> <td>d/cost</td> <td>-5</td> <td>5</td> <td>15</td> <td>45</td> <td>-55</td> <td>-35</td> </tr> <tr> <td>P</td> <td>E</td> <td>S</td> <td>A</td> <td>B</td> <td>C</td> <td>D</td> </tr> </table>		S	A	B	C	D	E	d/cost	0	Inf	Inf	Inf	Inf	Inf	P	Nil	Nil	Nil	Nil	Nil	Nil		S	A	B	C	D	E	d/cost	0	5	Inf	Inf	30	Inf	P	Nil	S	Nil	Nil	S	Nil		S	A	B	C	D	E	d/cost	-5	5	15	45	-55	-35	P	E	S	A	B	C	D	10	CO3	BL2
	S	A	B	C	D	E																																																													
d/cost	0	Inf	Inf	Inf	Inf	Inf																																																													
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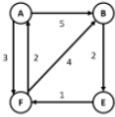
4.	Deduce the maximum flow for the network given below by using an approach which allows an intermediate node to store the excess flow locally.	10	CO3	BL3
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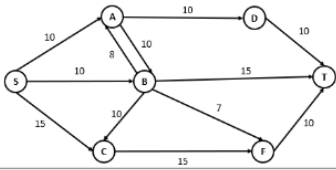
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The flow starts from the source, s, ends in the sink, t, and the values on the edges represent the corresponding edge capacities. Ans. Max Flow deduction by Push Relabel approach Initialization <table> <tr> <th></th><th>s</th><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>t</th></tr> <tr> <td>u.e</td><td>0/Inf</td><td>5</td><td>4</td><td>0</td><td>8</td><td>0</td><td>0</td></tr> <tr> <td>u.h</td><td>7</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr> </table> After relabel a (h=1), push from a to c and d, relabel a 7 times to a.h=8, push from a to s <table> <tr> <th></th><th>s</th><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>t</th></tr> <tr> <td>u.e</td><td>0/Inf</td><td>0</td><td>4</td><td>3</td><td>9</td><td>0</td><td>0</td></tr> <tr> <td>u.h</td><td>7</td><td>8</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr> </table> Relabel c (h=1), push from c to t <table> <tr> <th></th><th>s</th><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>t</th></tr> <tr> <td>u.e</td><td>0/Inf</td><td>0</td><td>4</td><td>0</td><td>9</td><td>0</td><td>3</td></tr> <tr> <td>u.h</td><td>7</td><td>8</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr> </table> Relabel b (h=1), push from b to c Relabel e (h=1), push from e to t <table> <tr> <th></th><th>s</th><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>t</th></tr> <tr> <td>u.e</td><td>0/Inf</td><td>0</td><td>0</td><td>0</td><td>9</td><td>0</td><td>7</td></tr> <tr> <td>u.h</td><td>7</td><td>8</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td></tr> </table> Relabel d (h=2), push from d to b, c, e Relabel c, e (h=2), push from c, e to t <table> <tr> <th></th><th>s</th><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>t</th></tr> <tr> <td>u.e</td><td>0/Inf</td><td>0</td><td>3</td><td>0</td><td>0</td><td>0</td><td>13</td></tr> <tr> <td>u.h</td><td>7</td><td>8</td><td>1</td><td>2</td><td>2</td><td>2</td><td>0</td></tr> </table> And so on ... Total u.e deposited at t would be 15, and all the internal nodes e to be zero.						s	a	b	c	d	e	t	u.e	0/Inf	5	4	0	8	0	0	u.h	7	0	0	0	0	0	0		s	a	b	c	d	e	t	u.e	0/Inf	0	4	3	9	0	0	u.h	7	8	0	0	0	0	0		s	a	b	c	d	e	t	u.e	0/Inf	0	4	0	9	0	3	u.h	7	8	0	1	0	0	0		s	a	b	c	d	e	t	u.e	0/Inf	0	0	0	9	0	7	u.h	7	8	1	1	0	1	0		s	a	b	c	d	e	t	u.e	0/Inf	0	3	0	0	0	13	u.h	7	8	1	2	2	2	0	10	CO3	BL3
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	s	a	b	c	d	e	t			
u.e	0/Inf	0	0	0	9	0	7			
u.h	7	8	1	1	0	1	0			
Relabel d (h=2), push from d to b, c, e										
Relabel c, e (h=2), push from c, e to t										
	s	a	b	c	d	e	t			
u.e	0/Inf	0	3	0	0	0	13			
u.h	7	8	1	2	2	2	0			
And so on ...										
Total u.e deposited at t would be 15, and all the internal nodes e to be zero.										
5.	Starting and end points of six line segments are given below. The objective is to find out if any of the line segments intersect or not by applying an algorithm which sweeps the xy coordinate space from left							10	CO3	BL3

	to right by a vertical line. Execution of the necessary intermediate steps is required. Line segment A (3,4) to (9,5), B (5,9) to (23,9), C (7,9) to (19,15), D (22, 52) to (25,54), E (14,6) to (29,14) F(10,11) to (15,7)			
	Ans. Line sweep technique There will be 12 event points , and they are : X=3 (Line A start), X=5 (Line B start), X=7 (Line C start), X=9 (Line A end), X=10 (Line F start), X=14 (Line E start), X=15 (Line F end), X=19 (Line C end), X=22 (Line D start), X=23 (Line B end), X=25 (Line D end), X=29 (Line E end).			
	1 st event point (x=3) only line A enters			
	2 nd Event point (x=5), line B enters, line B > line A. Does line B (5,9 to 23,9) and line A (3,4 to 9,5) intersect ? No . (Use two line segment intersection algorithm (time complexity O(constant) to get the answer).			
	3 rd event point (x=7), line C enters, C>B, does C & B intersect? Yes (Use two line segment intersection algorithm (time complexity O(constant) to get the answer).			

Q.No	Question	Max Mark															
1	Solve the Knapsack Problem using FIFOBB, assume knapsack capacity is $W=8$. Show how queue is used for node creation in the state space tree. <table border="1"> <thead> <tr> <th>Item</th><th>Profit</th><th>Weight</th></tr> </thead> <tbody> <tr> <td>1</td><td>13</td><td>4</td></tr> <tr> <td>2</td><td>15</td><td>2</td></tr> <tr> <td>3</td><td>14</td><td>4</td></tr> <tr> <td>4</td><td>16</td><td>6</td></tr> </tbody> </table>	Item	Profit	Weight	1	13	4	2	15	2	3	14	4	4	16	6	10
Item	Profit	Weight															
1	13	4															
2	15	2															
3	14	4															
4	16	6															
2	Find the existence of a pattern P in the given string S (assign digits A-C as 0-2), using Rabin Karp algorithm. For hash function use Mod 13. Find out how many spurious hits does the algorithm encounter in the Text = ABCBBCABCBAABCCACB when looking for the pattern Pattern = CCA?	10															
3	Consider a logistics manager tasked with optimizing transportation routes for a delivery company that operates in a city with a complex network of roads. Your goal is to minimize the distance for packages to reach their destinations by finding the shortest paths between all pairs of locations. The transportation route is represented as weighted directed graph given below. Find the shortest paths between all pairs of locations, considering the varying distance, which helps company to delivery operation. 	10															

4	In water distribution systems, we need to find the maximum amount of water that can be supplied from source S to destination T through a network pipes with capacity limitations. Given a directed graph $G=(V,E)$ representing a water distribution system, where V is set of vertices and E is the set of edges, each edge (u,v) has a capacity $c(u,v)$ representing the maximum water flow that can be supplied through the network pipe. Use Push Relabel algorithm to find the maximum water flow that can be supplied from node S to node T using given graph. 	10
5	Find whether the following line segments intersect or not using cross product. - L1 :{(1,23) & (10,15)} and L2 :{(4,10) & (6,20)} - L3 :{(4,5) & (7,10)} and L4 :{(1,1) & (5,5)} - L5 :{(1,1) & (10,10)} and L6 :{(3,3) & (5,5)} - L7 :{(1,1) & (10,10)} and L8 :{(5,8) & (3,3)}	10

1.	Consider the following items that can be added to the knapsack with a capacity of 16 kgs. Solve and obtain the optimal solution using the branch-and-bound FIFO method. Use variable and fixed representation to create the state space tree and differentiate between the representations. Note: Variable representation refers to the binary variables (0 or 1) that determine whether an item is included or excluded from the knapsack, while fixed representation refers to the predetermined, unchanging data associated with each item, such as its weight and value. <table><tr><td>Item</td><td>I1</td><td>I2</td><td>I3</td><td>I4</td></tr><tr><td>Profit</td><td>\$40</td><td>\$30</td><td>\$50</td><td>\$10</td></tr><tr><td>Weight (in kg)</td><td>2</td><td>5</td><td>10</td><td>5</td></tr></table>	Item	I1	I2	I3	I4	Profit	\$40	\$30	\$50	\$10	Weight (in kg)	2	5	10	5	2	3
Item	I1	I2	I3	I4														
Profit	\$40	\$30	\$50	\$10														
Weight (in kg)	2	5	10	5														
			10															



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SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025

REG.NO.:

SLOT: A2+TA2

	$W = 16$ Items selected : I1 and I3.		
2.	Consider the following gene sequence: AGCTTAGCTGAGCTAGCTAGCTGAGCTGCTAG where Adenine (A), Cytosine (C), Guanine (G), and Thymine (T) are the four bases in DNA. During analysis, assume that the following pattern often occurs: GCTAG	3	3

2. Consider the following gene sequence:

AGCTTAGCTGAGCTAGCTAGCTGAGCTGCTAG

where Adenine (A), Cytosine (C), Guanine (G), and Thymine (T) are the four bases in DNA. During analysis, assume that the following pattern often occurs: GCTAG (in HIV disease) and GAGCTG (in lung cancer disease) in the sequence. Identify the disease sequence that occurs more frequently by using the KMP string matching technique. Make the LPS table for the two patterns that are being considered. Give the steps for building the table.

Answer Key:

LPS table - 1 mark each

Finding the pattern using KMP string matching algorithm - 4 marks each

10

Failure function:

j	0	1	2	3	4
P[j]	G	C	T	A	G
f[j]	0	0	0	0	1

AGCTTAGCTGAGCTAGCTAGCTGAGCTGCTAG

Pattern GCTAG occurs twice

Page 2 of 5



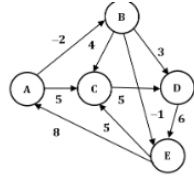
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REG.NO.:

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025

SLOT: A2+TA2

	Failure function: <table border="1"> <tr> <td>j</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr> <tr> <td>P[j]</td><td>G</td><td>A</td><td>G</td><td>C</td><td>T</td><td>G</td></tr> <tr> <td>f[j]</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td></tr> </table> AGCTTAGCTGAGCTAGCTAGCTGAGCTGCTAG Pattern GAGCTG occurs once Hence, the HIV disease sequence occurs often. Step by step visualization can be done at: https://cmps-people.ok.ubc.ca/yliucet/DS/KnuthMorrisPratt.html	j	0	1	2	3	4	5	P[j]	G	A	G	C	T	G	f[j]	0	0	1	0	0	1		
j	0	1	2	3	4	5																		
P[j]	G	A	G	C	T	G																		
f[j]	0	0	1	0	0	1																		
3.	Given a weighted directed graph, apply the Floyd-Warshall algorithm to compute the shortest paths between all pairs of nodes. After obtaining the distance matrix, construct the predecessor matrix to store the intermediate nodes along the shortest paths. Using this predecessor matrix, determine the paths for the following node pairs: a) From node A to node D b) From node D to node B	3	3																					



A

Final Answer:

```
vertex_labels = ["A", "B", "C", "D", "E"]
shortest_paths = [
    [0, -2, 2, 1, -3],
    [7, 0, 4, 3, -1],
    [19, 17, 0, 5, 11],
    [14, 12, 11, 0, 6],
    [8, 6, 5, 3, 0]
]
```

Predecessor Matrix



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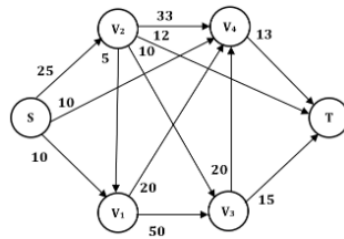
SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025

SLOT: A2+TA2

0 A B B B
E 0 B B B
E A 0 C D
E A E 0 D
E A E B 0

Path from A to D: A->B->D
Total path cost: 1
Path from D to B: D->E->A->B
Total path cost: 12

4. For the given graph that represents a water treatment plant with various stations, calculate the total network flow of the plant using the Ford Fulkerson algorithm. For every augmenting path selected, the residual graph must be clearly drawn.



Final max flow: 40

10

3 3

				
5.	From the given set of points, $\{(6, 1), (3, 3), (7, 5), (1, 2), (0, 5)\}$, create possible line segments so that the points are not repeated. Calculate d_1 , d_2 , d_3 , and d_4 for the	10	3	2

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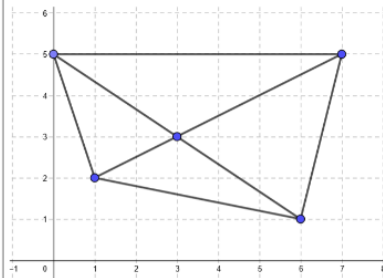
REG.NO.:

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025

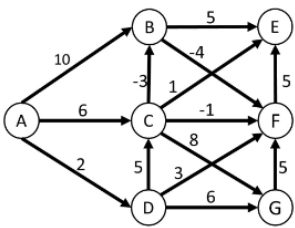
SLOT: A2+TA2

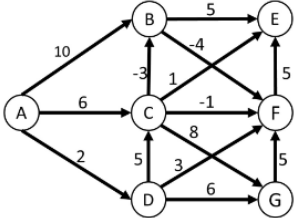
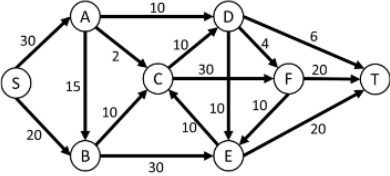
created line segments and check according to SEGMENT_INTERSECT procedure if they straddle each other or are collinear. Every step in the intersection process shall be shown.

Set of points	directions d_1, d_2, d_3, d_4		
(6,1) (3,3) (7,5)	(1,2) 1 2 2 0 onsegment	Collinear	Yes
(6,1) (7,5) (3,3)	(1,2) 2 2 2 0 No		
(6,1) (1,2) (3,3)	(7,5) 1 1 1 0 No		
Next Set of points	directions d_1, d_2, d_3, d_4		
(6,1) (3,3) (7,5)	(0,5) 1 0 2 2 No		
(6,1) (7,5) (3,3)	(0,5) 2 2 0 1 No		
(6,1) (0,5) (3,3)	(7,5) 0 1 1 2 onsegment	Collinear	Yes
Next Set of points	directions d_1, d_2, d_3, d_4		
(7,5) (3,3) (1,2)	(0,5) 0 1 1 1 No		
(7,5) (1,2) (3,3)	(0,5) 0 1 1 2 onsegment	Collinear	Yes
(7,5) (0,5) (3,3)	(1,2) 2 2 0 1 No		
Next Set of points	directions d_1, d_2, d_3, d_4		
(7,5) (6,1) (1,2)	(0,5) 1 1 1 1 No		
(7,5) (1,2) (6,1)	(0,5) 2 1 1 2 Yes		
(7,5) (0,5) (6,1)	(1,2) 2 2 1 1 No		
Next Set of points	directions d_1, d_2, d_3, d_4		
(3,3) (6,1) (1,2)	(0,5) 1 0 1 1 No		
(3,3) (1,2) (6,1)	(0,5) 2 1 0 2 onsegment	Collinear	Yes
(3,3) (0,5) (6,1)	(1,2) 0 2 1 1 No		

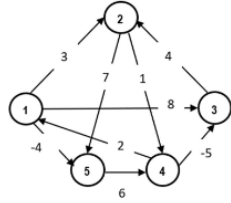
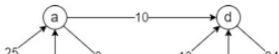


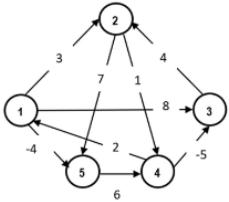
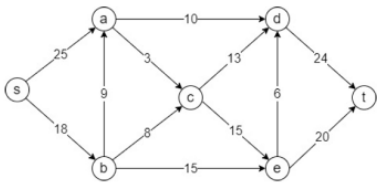
1	What is the difference between FIFOBB and LIFOBB? Solve the Knapsack Problem using LIFOBB, Assume knapsack capacity is $W = 8$. Show how stack is used for node creation in the state space tree using variable tuple representation. <table><tr><th>Item</th><th>Profit</th><th>Weight</th></tr><tr><td>1</td><td>3</td><td>3</td></tr><tr><td>2</td><td>4</td><td>2</td></tr><tr><td>3</td><td>5</td><td>3</td></tr><tr><td>4</td><td>6</td><td>5</td></tr></table>	Item	Profit	Weight	1	3	3	2	4	2	3	5	3	4	6	5	2	10					
Item	Profit	Weight																					
1	3	3																					
2	4	2																					
3	5	3																					
4	6	5																					
2	Solve the Job Sequencing Problem using FIFOBB. Select optimal subset J with an optimal penalty for the following data. What will be the penalty corresponding to the optimal solution? Discuss how queue is used in the state space tree using fixed tuple representation. <table><tr><th>Job</th><th>Penalty</th><th>Deadline Time</th><th>Execution Time</th></tr><tr><td>1</td><td>40</td><td>2</td><td>1</td></tr><tr><td>2</td><td>10</td><td>4</td><td>2</td></tr><tr><td>3</td><td>60</td><td>2</td><td>2</td></tr><tr><td>4</td><td>30</td><td>3</td><td>1</td></tr></table>	Job	Penalty	Deadline Time	Execution Time	1	40	2	1	2	10	4	2	3	60	2	2	4	30	3	1	2	10
Job	Penalty	Deadline Time	Execution Time																				
1	40	2	1																				
2	10	4	2																				
3	60	2	2																				
4	30	3	1																				
3	Using a string matching algorithm which uses hashing functions, how many spurious hits does the algorithm (with q to be 11) encounter in the Text = ABCBCABCBA when looking for the pattern Pattern = CBA?	3	10																				

4	John is driving his electric car from Vellore and has planned to visit tourist places around Vellore. Due to limitations of charging on the road, he needs to find the shortest path from Vellore to all other chosen tourist places, through. The road map with tourist places between Vellore and tourist places is given as a weighted directed graph $G = (V,E)$ along with weights w (representing the distance). Find the shortest path from source vertex A (i.e., Vellore) to all other vertices (i.e., to all other visiting place around Vellore) using suitable algorithm for the given graph. Write the pseudocode of your algorithm. 	4	10
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5	In a wireless network, node S wishes to send some packets to node T, but S and T are not directly connected. We need to find the path that can transfer the most packets from node S to node T, while satisfying the capacity constraints of the edges. The network is represented as a graph, $G = (V,E)$. Design an algorithm with complexity $O(V E^2)$, and find the maximum number of packets transferred from node S to node T using the given graph. 	4	10

Q.No	Question	max Marks	CO	BL																		
1.	Differentiate between FIFO and LIFO batch and bound methods for the following 0-1 Knapsack problem where knapsack capacity is 15 kg: <table><tr><td>Item</td><td>X1</td><td>X2</td><td>X3</td><td>X4</td><td>X5</td></tr><tr><td>Profit</td><td>5</td><td>7</td><td>9</td><td>10</td><td>12</td></tr><tr><td>Weight (in kg)</td><td>2</td><td>5</td><td>7</td><td>8</td><td>10</td></tr></table> Illustrate the FIFO and LIFO-based knapsack solution trees.	Item	X1	X2	X3	X4	X5	Profit	5	7	9	10	12	Weight (in kg)	2	5	7	8	10	10	CO 2	BL 5
Item	X1	X2	X3	X4	X5																	
Profit	5	7	9	10	12																	
Weight (in kg)	2	5	7	8	10																	
2.	Compare the naive string-matching algorithm with the KMP string matching algorithm. What are the drawbacks of the naive string matching algorithm that are overcome by the KMP algorithm? Apply both Naive and KMP algorithms for the following: <i>String: apphappyapihappiness</i> <i>Pattern: appi</i>	10	CO 3	BL 4																		

3.	Find the existence of a pattern P in the given string S (using digits 0-9), using Rabin Karp algorithm. S = "3124182284769678922843256" P = "2284" For the hash function, use mod value 997.	10	CO 3	BL 4
4.	Apply the Floyd-Warshall algorithm to the following graph. Show the values after each iteration. List out the final shortest paths. 	10	CO 3	BL 4
5.	Use the Edmond Karp method to determine the maximum flow possible in the given flow network. Adequate explanations and neat diagrams are expected in each step. 	10	CO 3	BL 3

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